

Article

Analyzing an Extreme Rainfall Event in Himachal Pradesh, India, to Contribute to Sustainable Development

Nitin Lohan ^{1,2}, Sushil Kumar ¹, Vivek Singh ³, Raj Pritam Gupta ⁴ and Gaurav Tiwari ^{5,*} 

¹ Department of Applied Mathematics, Gautam Buddha University, Greater Noida 201312, India; lohannitin@gmail.com (N.L.); sushil12@gmail.com (S.K.)

² India Meteorological Department, Ministry of Earth Sciences (MoES), New Delhi 110003, India

³ Indian Institute of Tropical Meteorology, (New Delhi Branch), Ministry of Earth Sciences (MoES), New Delhi 110060, India; vivek.sgh1@gmail.com

⁴ Department of Earth and Environmental Sciences, Indian Institute of Science Education and Research Bhopal, Bhauri 462066, India; raj23@iiserb.ac.in

⁵ Center for Environmental Remote Sensing, Chiba University, Chiba 2630022, Japan

* Correspondence: gtiwari506@gmail.com

Abstract: In the Himalayan regions of complex terrains, such as Himachal Pradesh, the occurrence of extreme rainfall events (EREs) has been increasing, triggering landslides and flash floods. Investigating the dynamics and precipitation characteristics and improving the prediction of such events are crucial and could play a vital role in contributing to sustainable development in the region. This study employs a high-resolution numerical weather prediction framework, the weather research and forecasting (WRF) model, to deeply investigate an ERE which occurred between 8 July and 13 July 2023. This ERE caused catastrophic floods in the Mandi and Kullu districts of Himachal Pradesh. The WRF model was configured with nested domains of 12 km and 4 km horizontal grid resolutions, and the results were compared with global high-resolution precipitation products and the fifth-generation European Centre for Medium-Range Weather Forecasts atmospheric reanalysis dataset. The selected case study was amplified by the synoptic scale features associated with the position and intensity of the monsoon trough, including mesoscale processes like orographic lifting. The presence of a western disturbance and the heavy moisture transported from the Arabian Sea and the Bay of Bengal both intensified this event. The model has effectively captured the spatial distribution and large-scale dynamics of the phenomenon, demonstrating the importance of high-resolution numerical modeling in accurately simulating localized EREs. Statistical evaluation revealed that the WRF model overestimated extreme rainfall intensity, with the root mean square error reaching 17.33 mm, particularly during the convective peak phase. The findings shed light on the value of high-resolution modeling in capturing localized EREs and offer suggestions for enhancing disaster management and flood forecasting.

Keywords: Himalayan region; extreme rainfall events; WRF model; flash floods



Academic Editor: Hossein Bonakdari

Received: 1 February 2025

Revised: 24 February 2025

Accepted: 24 February 2025

Published: 28 February 2025

Citation: Lohan, N.; Kumar, S.; Singh, V.; Gupta, R.P.; Tiwari, G. Analyzing an Extreme Rainfall Event in Himachal Pradesh, India, to Contribute to Sustainable Development.

Sustainability **2025**, *17*, 2115. <https://doi.org/10.3390/su17052115>

Copyright: © 2025 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license

(<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Extreme rainfall events (EREs) have been significantly on the rise within the last few decades and are producing increasingly worse problems worldwide [1–5]. In most cases, such EREs have devastating impacts ranging from floods and landslides to the extensive damage of structures and the loss of life. According to the historical trend, the frequency of heavy precipitation events has increased over the past century [3,6–10]. It is indicated that if global temperature increases to 2 °C, the severity of extreme rainfall which may occur

once in a decade or in half a century could rise by about 15% [11]. The most contributory reason behind this rise in EREs is quick climate change that favors more water retention in the atmosphere and finally leads to rainfall that is usually much more extensive [12–14]. Other regions that show an increasing trend in heavy rainfall events are South Asia, Europe, and North America, particularly in complex terrains [15–17].

In recent times, there has been a notable increase in EREs over the Indian subcontinent, which is largely affected by the Indian summer monsoon (ISM) [10,18]. According to several studies, there has been little to no rise in the average seasonal monsoon rainfall, but there has been an increase in the frequency and intensity of short-duration heavy rainfall events [5,19]. This shift is particularly concerning as it disrupts water resource management strategies, increases the risk of flooding, and affects agricultural economies [19,20]. The ISM, which occurs from June to September, is responsible for nearly 80% of India's annual rainfall [18,21,22]. It is influenced by various large-scale circulation systems, such as the monsoon trough, low-pressure systems, western disturbances (WDs), Madden–Julian Oscillation, El Niño–Southern Oscillation (ENSO), and Indian Ocean Dipole [23,24]. The Arabian Sea and the Bay of Bengal are significant sources of moisture for this whole monsoon season. A key factor contributing to the increase in EREs in Northern India, especially in the Western Himalayan region, is the interaction between monsoonal systems and extratropical disturbances, i.e., WDs [25,26]. Recent studies have shown that the coexistence and temporary merging of monsoon lows and WDs can result in extremely heavy rainfall events [27,28]. Monsoon lows, which are low-pressure systems, form within a monsoon region during the monsoon season (June–September). On the other hand, WDs originate in the Mediterranean region and bring winter rain to the northwestern parts of the Indian subcontinent. This interaction is often accompanied by Rossby wave breaking, which enhances upper-level divergence and intensifies precipitation in the area [29,30]. In the Rossby wave breaking phenomenon, Rossby waves undergo a transition from smooth and undisturbed to a turbulent state, which alters the weather patterns and climate processes.

The Himalayan region is highly vulnerable to EREs due to its complex terrain, unique climatic conditions, and proximity to significant monsoon systems [8,31–33]. The Western Himalayas, covering Himachal Pradesh, Uttarakhand, and Jammu and Kashmir, frequently experience extreme rainfall and cloudburst events. These intense weather phenomena are primarily driven by strong orographic lifting and the interaction between large-scale and mesoscale weather systems [20]. In recent years, several EREs have caused severe destruction in this area. For instance, the June 2013 Kedarnath disaster in Uttarakhand was triggered by heavy rainfall and devastating flash floods that resulted in thousands of casualties and widespread destruction [34,35]. Similarly, a cloudburst in Himachal Pradesh in July 2021 caused massive flooding and landslides, underscoring growing concerns about the increasing frequency and impact of EREs in the region [15]. Numerous studies have connected climate change in mountainous areas to an increase in the frequency of extreme occurrences [36,37]. The rise in tropospheric water vapor content over the Indian subcontinent was emphasized by [38]. The study of the Himalayan region has also benefited from the rising significance of satellite records. Satellite-based nowcasting techniques have particularly enhanced early warning systems by precisely forecasting times of heavy precipitation across the Western Himalayan region [39]. Similarly, Shah et al., (2023) [40] verified satellite-based cloudburst alerts, demonstrating exceptional location precision over Uttarakhand.

In recent years, the Weather Research and Forecasting (WRF) model [41] has been widely used for mesoscale weather phenomenon studies and operational weather prediction at various time scales globally [42–50]. The WRF model simulates a variety of high-impact weather events such as precipitation, heatwaves, thunderstorms, and trop-

ical cyclones [42,48,51–62]. Chevuturi et al. (2015) [63] simulated an ERE in the Central Himalayas, while Kumar et al. (2012) [64] studied the atmospheric processes that led to the 2010 Leh cloudburst event. Studies by Boyaj et al. (2020) [51] suggested that land use and land cover change caused by rapid urbanization can enhance rainfall by >20% during heavy rainfall events. Many more researchers have employed the WRF model to investigate different case studies over the Himalayan region [65–68]. In July 2023, very severe rainfall episodes occurred in Himachal Pradesh (India), where some areas recorded a staggering 436% increase in rainfall compared to the average. Only a few studies have conducted a comprehensive statistical evaluation of high-resolution WRF simulations against multiple observational datasets, especially using satellite-based precipitation products for a more robust validation. Thus, we have employed the WRF model for this extreme rainfall event to evaluate the model's performance at fine horizontal grid resolution and to analyze the synoptic and mesoscale features that contribute to this event. In this way, the study addresses the importance of numerical modeling for sustainable development with regard to ERE forecasting.

2. Credentials of Extreme Rainfall Event from Satellite Product

As per the India Meteorological Department's (IMD) reports, many parts of Northern India covering the states of Himachal Pradesh, Uttarakhand, Punjab, Haryana, Chandigarh, and Delhi received unusually significant rainfall, with many stations reporting heavy (64.5–115.5 mm), very heavy (115.6–204.4 mm), and extremely heavy rainfall (≥ 204.5 mm) from 8 to 13 July 2023. Several locations in Himachal Pradesh recorded heavy rain during this period, with some regions receiving more than 200 mm of rainfall in a single day [7]. This ERE caused the widespread destruction of life and property over the region. As per the report, there were two active synoptic systems present at the same time. The western end of the monsoon trough at the mean sea level was very active and was to the south of its normal position from the 8th to the 12th morning (IMD report). Another synoptic system that was also active over Northwest India was the WD as a trough in the mid-troposphere. Between 8 and 13 July, both systems merged over the Himalayan region and ended with a heavy downpour over the Western Himalayas.

The Indian National Satellite-3D Repeat (INSAT 3DR) derived a brightness temperature (BT; Unit: Kelvin) analysis from 7 July to 12 July 2023, revealing the evolution of a deep convection associated with the ERE (Figure 1). Initially, on 7 July, widespread cloud cover was observed over Northwestern India, with scattered convection over Central India. By 8–9 July, the convective system intensified significantly and merged, with widespread low BT values over Himachal Pradesh, indicating a mesoscale convective system development. On 10 July, the convective system reached its peak intensity, covering most of Himachal Pradesh. After 10 July, the system became weak and started moving eastward.

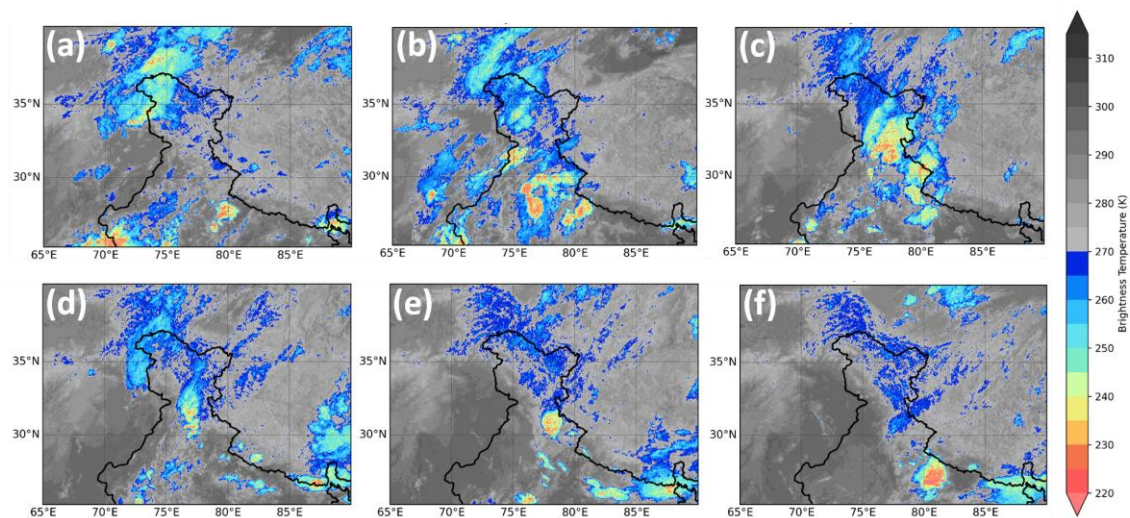


Figure 1. The daily evolution of infrared brightness temperature (Unit: Kelvin) was derived from the INSAT-3DR satellite product. Panel figures (a–f) are plotted from 8 to 13 July 2023, respectively.

3. Materials and Methods

The numerical simulation of the selected ERE was performed by using the fully compressible and non-hydrostatic Weather Research and Forecasting (WRF) model version 4.5.1. The WRF model was configured with the following nested domains: parent domain (D01) with a coarser horizontal grid resolution of 12 km and child domain (D02) with a finer horizontal grid resolution of 4 km (Figure 2). The initial and lateral boundary conditions were obtained from the National Centers for Environmental Prediction (NCEP) Final Analysis (FNL) dataset, available after 6 h from the real-time at a horizontal resolution of $0.25^\circ \times 0.25^\circ$. The simulations were initialized at UTC +00:00 on 08 July and integrated for 144 h, covering all phases of the ERE. A total of 272,410 and 478,502 grids of 12 km and 4 km, with 55 vertical levels and 30 s and 10 s of integration time steps, were used to cover the outer and inner domains, respectively. The WRF model used the Arakawa C grid system. Table 1 provides detailed information about the model setup and the chosen parameterization schemes.

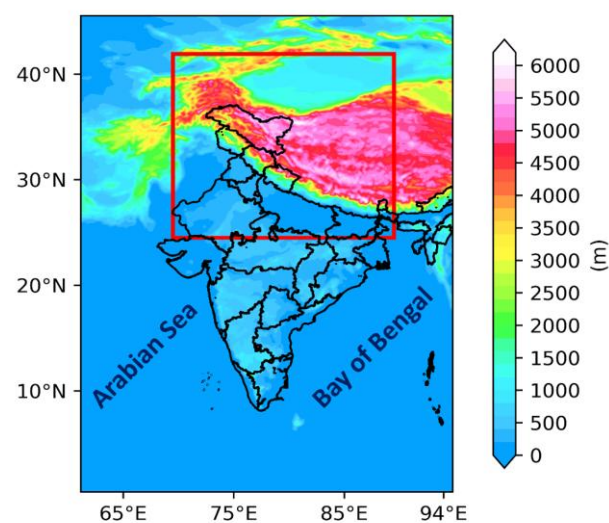


Figure 2. The plotted area of the figure demonstrates the dimensions of the outer domain (D01). A rectangular box indicates the dimensions of the inner domain (D02), along with the topography of the study domains.

Table 1. Description of WRF model’s domain setup and physical parameterization schemes.

Description	Outer Domain	Inner Domain
No. of Grid points in $X \times Y$	272, 410	478, 502
Domain Resolution	12 km	4 km
Vertical levels	55	55
Time Step	30 s	10 s
Duration of simulation	144 h	
Horizontal grid system	Arakawa C grid	
Acoustic and gravity wave model	3rd order Runge–Kutta scheme	
Longwave radiation	Rapid radiative transfer model (RRTM)	
Shortwave radiation	Dudhia scheme	
Surface Physics	Revised MM5 scheme	
Land surface	Unified Noah land surface model	
Planetary boundary layer scheme	Yonsei University scheme	
Cumulus parameterization scheme	Kain-Fritsch scheme	
Cloud microphysics scheme	WRF single-moment six-class	
Map projection	Mercator	

This study utilized multiple observational, satellite-based, and reanalysis products for various purposes (Table 2). The IMD’s gridded rainfall data were used to verify the rainfall patterns. ERA5 is a global atmospheric reanalysis product produced by the European Centre for Medium-Range Weather Forecasts (ECMWF), with a horizontal resolution of $0.25^\circ \times 0.25^\circ$ and 137 pressure levels from the earth’s surface to the top of the atmosphere. Multi-Source Weighted-Ensemble Precipitation (MSWEP) is a global precipitation data product. It uses multiple data sources (e.g., satellite measurements, ground-based observations, and reanalysis data). The 3-hourly MSWEP product covers a wide range of the years 1979 to the present.

Table 2. Brief description of the various datasets used in this study.

Data Source	Spatial Resolution	Purpose	Reference
IMD Gridded Rainfall	$0.25^\circ \times 0.25^\circ$	To verify daily spatial rainfall	[69]
ERA5	$0.25^\circ \times 0.25^\circ$	To verify large-scale features	[70]
MSWEP	$0.25^\circ \times 0.25^\circ$	For comparison with WRF rainfall	[71]
INSAT 3DR	4 km	For synoptic scale features	[72]
NCEP FNL	$0.25^\circ \times 0.25^\circ$	To prepare input forcings for the WRF model	[73]

3.1. Vertically Integrated Moisture Transport (VIMT)

Moisture transport plays a vital role in atmospheric circulation, weather phenomena, and precipitation. The VIMT represents the movement of water vapor content in the atmosphere over a vertical column [74]. Mathematically, it can be expressed as follows:

$$Q = \frac{1}{g} \int_{p_s}^{p_t} qVdp$$

where g , p_t , p_s , V , and q are the acceleration due to gravity, top of the atmosphere pressure, surface pressure, wind vector, and specific humidity, respectively. The vertical integration

of the equation is carried out from surface pressure level to 300 hPa, since specific humidity has a negligible effect above this level and is not a part of the reanalysis [75].

3.2. Equitable Threat Score (ETS)

The equitable threat score or Gilbert skill score evaluates the proficiency of a model in forecasting whether a specific weather event has occurred or not. A contingency table is used to compare the observed and forecasted values.

$$ETS = \frac{H - E}{H + M + F - E}$$

where H , M , and F denote the hits, misses, and false alarms respectively. E is the number of hits expected by chance.

3.3. Mean Bias Error (MBE) and Root Mean Square Error (RMSE)

The MBE quantifies the average tendency of the model to overestimate or underestimate rainfall compared to observations. A positive value of MBE indicates that the model overestimated the rainfall, whereas a negative MBE indicates that the model underestimated the rainfall. It is calculated as follows:

$$MBE = \frac{1}{N} \sum_{i=1}^N R_{WRF,i} - R_{OBS,i}$$

$RMSE$ measures the overall magnitude of error between the simulated and observed rainfall. Unlike MBE , it penalizes large deviations more heavily. It is calculated as follows:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (R_{WRF,i} - R_{OBS,i})^2}$$

where $R_{WRF,i}$ is the WRF simulated rainfall at grid point i , $R_{OBS,i}$ is the observed rainfall from MSWEP, and N is the total number of the grid points.

4. Results and Discussion

4.1. Rainfall Analysis

The 40-year climatological mean rainfall over the study region, valid for six days (8 to 13 July), is shown in Figure 3a. It is evident from Figure 3a that the magnitude of climatological mean is around 10–20 mm/day during this period over most of the Western Himalayan region. It reveals that the Himalayan foothills receive more rainfall than the mountaintop. Figure 3b shows the rainfall anomaly (mm) for the year 2023. During 8–13 July 2023, the Western Himalayas receives a higher amount of precipitation than the climatological mean. Southern Himachal Pradesh, Chandigarh, and some parts of Haryana receive 40 mm/day more rainfall than normal (Figure 3b). The large positive anomaly indicates the extraordinary nature of the heavy rainfall event. The highest positive anomalies are over the Kullu, Mandi, and Kangra districts of Himachal Pradesh, Chandigarh, and some parts of Haryana, indicating widespread flooding. The high positive anomalies align with low BT vales (shown in Figure 1), confirming a well-organized convective system over the region.

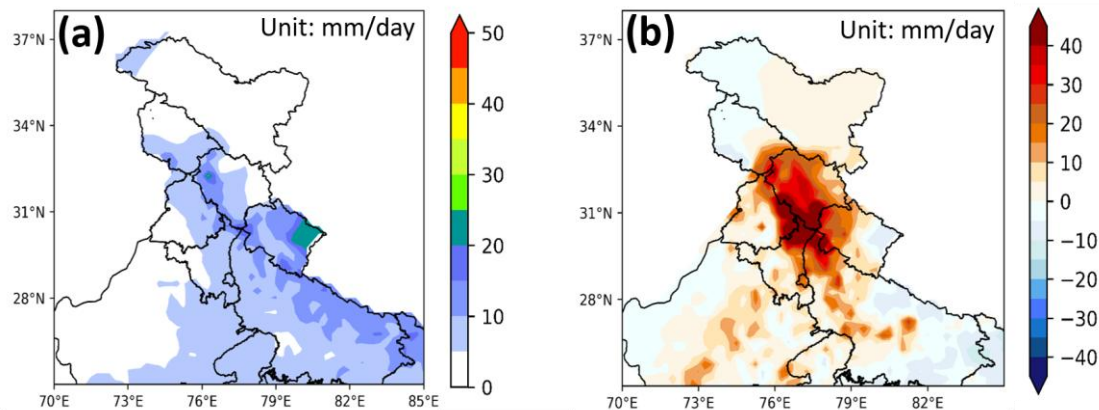


Figure 3. (a) Climatological mean rainfall distribution (mm/day) for the six days (8 July to 13 July) over the 40 years (1984 to 2023); (b) Rainfall anomaly for the period from 8 July to 13 July for 2023.

To investigate the spatiotemporal evolution of rainfall spread over the region, the daily rainbands were depicted for all six days using the IMD gridded data, ERA5, MSWEP products, and WRF model inner domain simulation's output (Figure 4). The model simulated the spatial distribution of rainfall in the initial days. As the system intensified, the model captured a marked increase in rainfall over Himachal Pradesh, particularly over the Kullu and Kangra districts, agreeing with ERA5, MSWEP and IMD. The captured rainfall also coincided with low BT values, suggesting the presence of a well-organized convective system. On 9 July, the system reached its peak intensity, with the maximum daily accumulated rainfall over Himachal Pradesh exceeding 140 mm. This phase marked the most active and widespread precipitation, influenced by the interaction of a monsoon low-pressure system and WD. Thus, the WRF model successfully captured the spatial and temporal evolution of rainfall, although some regions exhibited a slight overestimation. From 11 July onwards, the convective system started to weaken, showing a southward shift in the rain distribution (Figure 4). This shift in rainfall also coincided with the shift in low BT values (Figure 1).

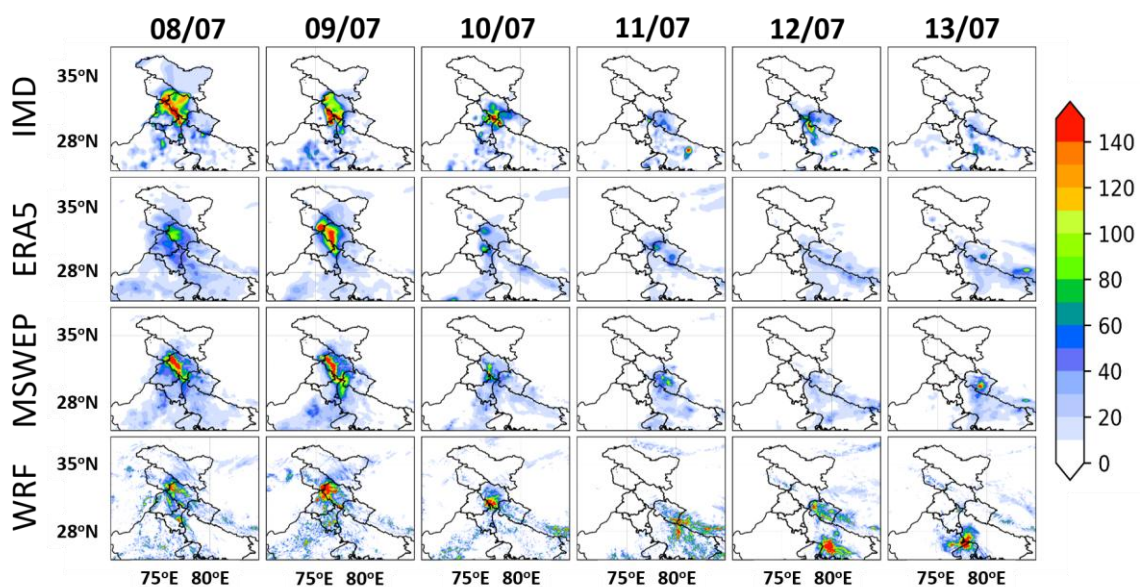


Figure 4. Spatiotemporal distribution of daily rainfall (mm) valid for six days (8 to 13 July 2023) from the IMD gridded data (top row), ERA5 (second row), MSWEP data (third row), and the WRF model's inner domain simulation (bottom row).

Furthermore, the MSWEP dataset was taken as the true value for the statistical verification of WRF rainfall due to its finer resolution. The equitable threat score for the inner domain's simulated rain for all six days at the various thresholds is shown in Figure 5. At the lower threshold (10–30 mm), the model exhibited a good ETS for 8, 9, 10, and 11 July (Figure 5). The maximum equitable threat score (0.35) was observed on 10 July. This indicates that the WRF model accurately simulated widespread rainfall during this period. However, for higher thresholds (>50 mm), the ETS declined (~below 0.2), highlighting that the model faced challenges in simulating localized extreme rainfall accurately. The decline was largely due to the underestimation of localized heavy rain. A day-to-day variability in the equitable threat score was present at the end of this HRE (Day 6) because the model needs additional components to simulate localized rain. This could be a good point of interest for future research.

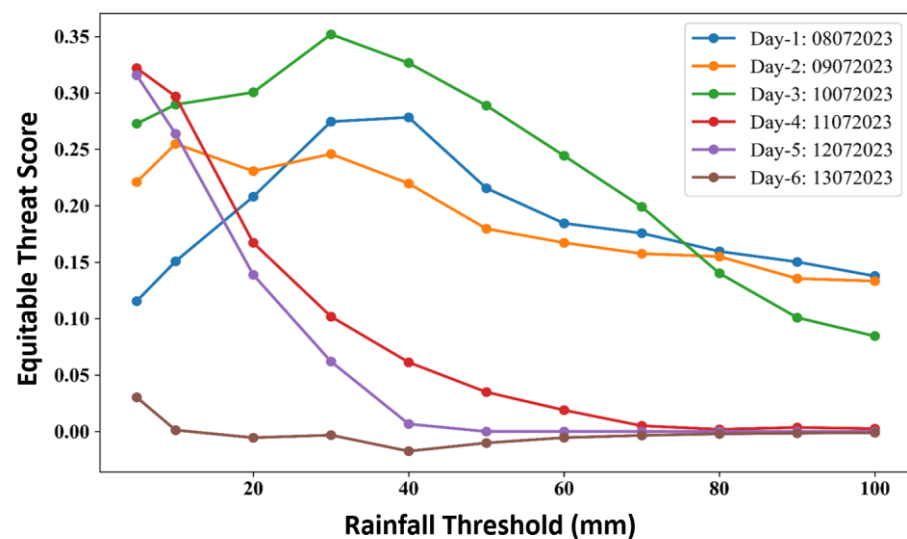


Figure 5. The Equitable Threat Score (ETS) for simulated rainfall (inner domain) validated against the MSWEP product at various threshold values from 8 July to 13 July 2023.

The MBE, correlation coefficient (CC), mean absolute error (MEA), and RMSE for each day of the ERE were calculated for the inner domain simulation against the MSWEP product (Table 3). The results indicated a progressive increase in the MBE values over the initial four days. Day 1 exhibited a near-zero bias (−0.13 mm), and this bias increased to a peak positive on day 4 (2.72 mm). This means that the model initially underestimated rainfall on day 1 but subsequently overestimated it as the event intensified, indicating that the model tended to simulate higher rainfall compared to the MSWEP during the peak convective phase of the event. Also, the CC values exhibited a steady trend of decline throughout the six days, ranging from 0.65 to 0.18. This indicates that the mode captured the rainfall variability better in the early phase of the event but struggled in the later phase. This declining trend is likely attributed to shifts in the large-scale forcing mechanisms and the weakening of organized rainfall systems.

The RMSE and MAE trends further reinforced these findings. The RMSE values were highest on days 2, 4, and 5 of the event, indicating large deviations between the WRF and the MSWEP during the maximum rainfall period. The MAE remained relatively stable during all days of the event. The low CC values with high RMSE and MBE values in the last phase of the event indicate that the overestimation of extreme rainfall also contributed to spatial mismatches in the rainfall distribution. The combined spatial and statistical analysis suggests that although the model effectively simulated this ERE over Himachal Pradesh,

the systematic overestimation of extreme rainfall intensity and localized discrepancies with the MSWEP data highlight the areas for potential improvement.

Table 3. The day-wise mean bias error, correlation coefficient, mean absolute error, and root mean square error for the simulated rain (inner domain) against the MSWEP data.

Day	MBE (mm)	Correlation Coefficient	MEA	RMSE
08/07/2023	−0.13	0.65	6.70	13.84
09/07/2023	1.02	0.59	6.94	16.81
10/07/2023	1.9	0.56	5.99	14.74
11/07/2023	2.72	0.46	6.47	16.57
12/07/2023	2.5	0.31	5.94	17.33
13/07/2023	0.5	0.18	5.23	15.41

4.2. Vertically Integrated Moisture Transport

As mentioned in Section 3.2, the role of VIMT is very crucial in determining the spatiotemporal evolution of extreme rainfall events, as it governs the large-scale movement of moisture from the source region to the area of heavy rainfall. Figure 6 provides the VIMT distribution in the outer domain for all six individual days using the ERA5 data. It can be observed a strong south-westerly moisture flux from the Arabian Sea, intensifying from 8 July onwards. The peak moisture transport is observed on 9 July, coinciding with the highest recorded rainfall in the region. This moisture transport aligns with the active monsoon phase. The intense northward advection of moisture, particularly along the monsoon trough axis, suggests the presence of a well-organized monsoon system. This monsoon system interacts with orographic features and a WD, leading to heavy rainfall over the region.

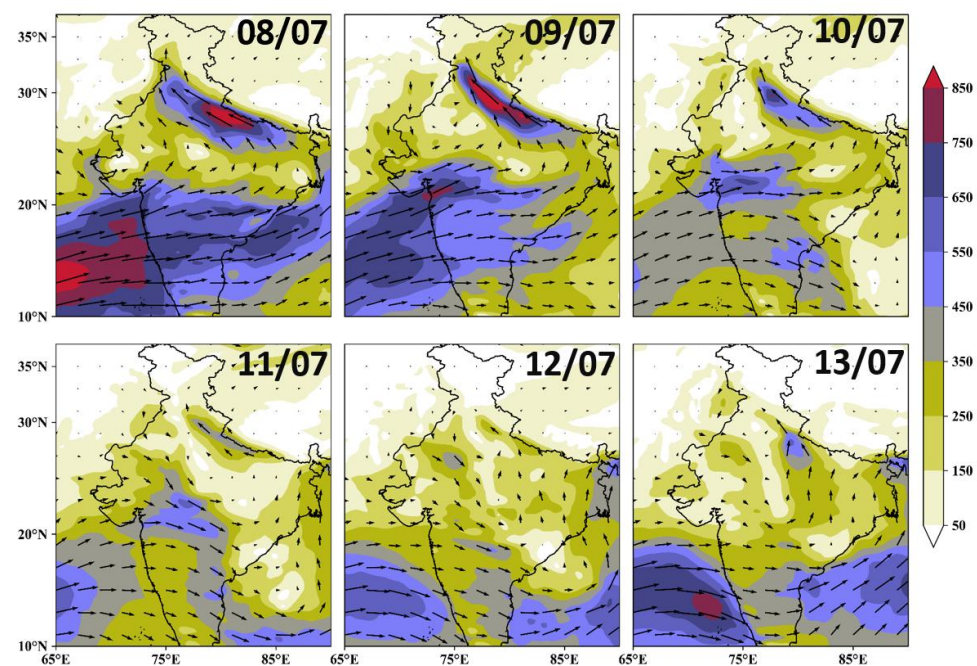


Figure 6. Vertically integrated moisture transport (VIMT; $\text{kg.m}^{-1}.\text{s}^{-1}$) for all six days from the ERA5 data. The contours are presenting the VIMT and vectors denote the flow of moisture transport.

To weigh the aptitude of the WRF model on detecting moisture influx and the associated dynamics leading to an ERE over Himachal Pradesh, the VIMT was calculated

from the model simulation that was valid for the outer domain (Figure 7). The model perceived broad moisture transport patterns similar to Figure 6 and captured the strong south-westerly inflows from the Arabian Sea and moisture convergence over the western Himalayan foothills. However, the model showed a slightly higher localized moisture flux over the Himalayan region, particularly on 9 July. The moisture vectors were also more pronounced in the WRF analysis. This suggests that the model simulated large-scale monsoonal moisture transport realistically. From 11 July, both the WRF and ERA5 exhibited a south-eastward retreat of moisture transport, indicating a weakening of this event. However, slight differences between ERA5 and model simulation remain, with the WRF indicating more concentrated moisture flux near the foothills, while ERA5 shows more widespread moisture transport over the Arabian Sea. The overall analysis highlights the critical role of moisture transport in driving this ERE. Also, this analysis demonstrates the WRF model's potential to successfully reproduce the monsoon-driven moisture influx comparable to ERA5, albeit with slightly stronger local convergence.

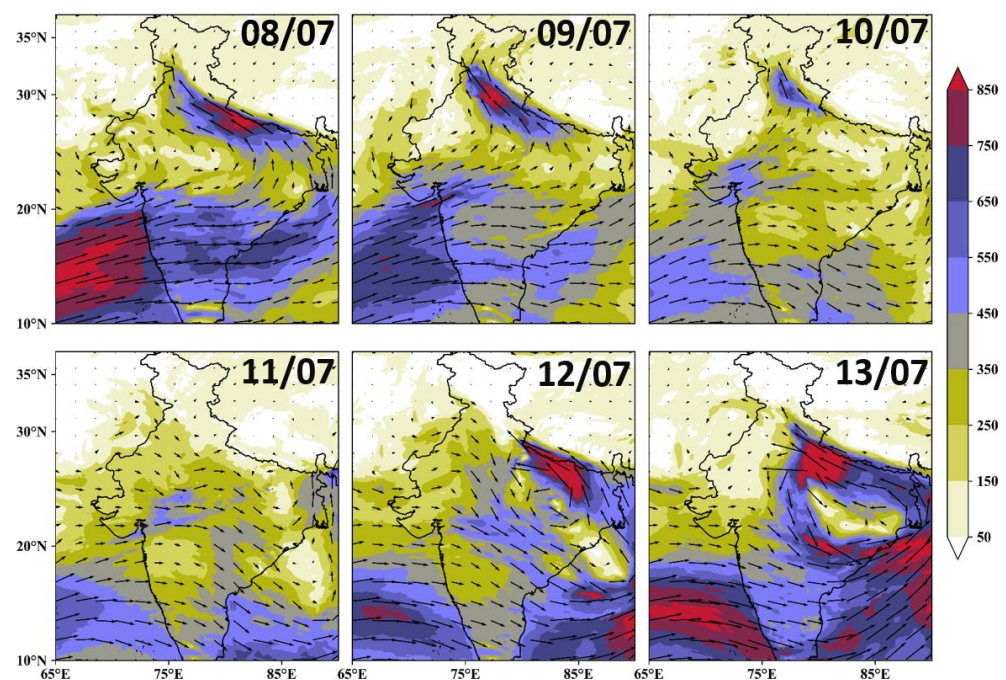


Figure 7. Vertically integrated moisture transport (VIMT; $\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-1}$) for all six days from the WRF model simulation.

4.3. Vertical Profile of Relative Humidity

The localized or small-scale dynamics of the extreme rainfall event are investigated using area-averaged time versus pressure levels cross-sections for relative humidity (RH; Figure 8). The area average is taken over Himachal Pradesh (30°N – 33.5°N and 75.5°E – 79°E). The vertical cross-section allows us to examine the depth of moisture availability, its vertical transport, and the efficiency of convection on both datasets. The ERA5 dataset (Figure 8a) highlights high RH ($>85\%$) across all pressure levels on the peak days of ERE, 8 July and 9 July. This suggests the presence of a deeply saturated column, indicating a strong convective activity and effective vertical transport. The most intense RH is between 800 hPa and 500 hPa, highlighting the role of monsoon dynamics and large-scale moisture convergence in sustaining deep convection. The rapid increase in RH over the Himalayan region often leads to cloudburst [76]. During the later phase (11–13 July), a noticeable drying trend is observed in the mid- and upper troposphere, suggesting a weakening of

this ERE. This RH trend aligns well with the decreasing rainfall intensity observed in the MSWEP and IMD datasets.

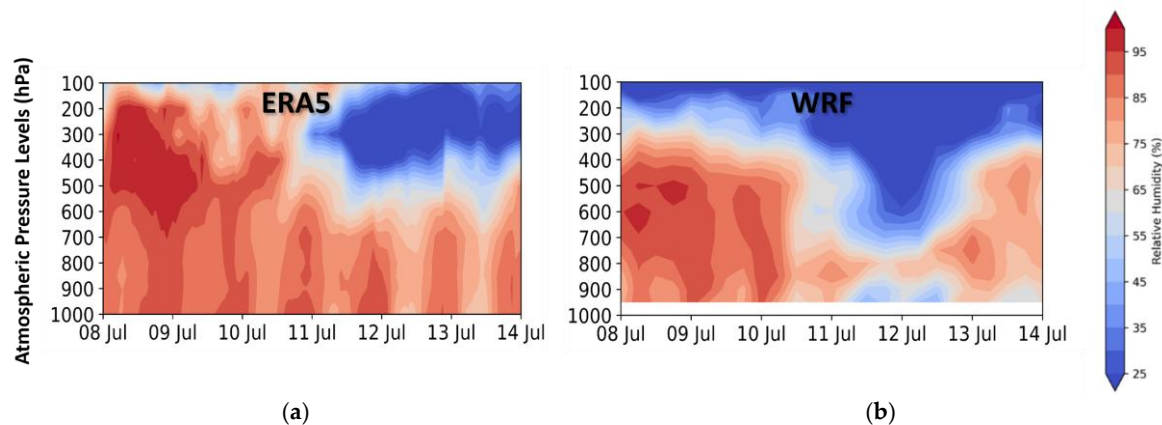


Figure 8. Area-averaged pressure vs. time vertical distribution of relative humidity (%) from (a) ERA5 and (b) WRF simulation for the inner domain.

The WRF inner domain's simulated RH cross-section (Figure 8b) captures a broadly similar RH profile, with high RH values during the ERE in the lower and mid-troposphere. However, WRF captures a slightly shallower moist layer. Also, the model-simulated RH values decline more rapidly above 500 hPa, suggesting that the model may be underestimating the depth of convection. Localized pockets of high RH are well captured by WRF. In contrast to ERA5, WRF exhibits slightly drier conditions in the upper troposphere even during the peak phase, which may indicate a weaker representation of the deep convective system and cloud microphysics compared to ERA5. From July 12 onwards, both datasets indicate a significant reduction in the RH. However, WRF maintains a slightly higher RH value in lower levels compared to ERA5. This suggests a slower drying-out process in the boundary layer. Overall, the WRF effectively captures RH variability and saturation trends but exhibits a notable difference in the upper-level RH representation compared to ERA5. The slightly shallower moist layer and localized RH maxima suggest a model adjustment in parameterization and cloud microphysics.

4.4. 700 hPa Daily Geopotential Height and Wind Pattern

The daily geopotential height (shaded; unit: m) and wind flow (vectors; unit: m/s) from ERA5 and WRF outer domain's simulations at 700 hPa are shown in Figure 9 to provide valuable insights into the atmospheric conditions conducive to the ERE. In general, the simulated geopotential height is slightly lower than the ERA5 counterpart for all days, but the contours are closely matchable. These marginally lower geopotential heights show the model's tendency for a slightly deeper low-pressure system and a more pronounced trough. For all six days, including the day of peak rainfall intensity (10 July), the reasonably comparable wind flows from ERA5 and WRF tend to lead to the accumulation of moisture in the lower troposphere, which can enhance the potential for cloud formation and heavy rains.

4.5. Applications of High-Resolution NWP Models in EREs Disaster Risk Reduction

Extreme rain disaster preparedness is demonstrated using a block diagram in Figure 10. High-resolution NWP models are pivotal in disaster risk reduction. Reliable and appropriate data are fundamental to comprehending the behavior and progression of EREs. These comprise weather station observations, satellite imagery, and radar observations. To assess the possible effects and direct decision-making, sophisticated computational

models analyze the data and generate probabilistic forecasts. For a timely and efficient response, forecast information must be communicated to the public and the authorities. All stakeholders should receive warnings and advisories through efficient communication strategies that utilize multiple channels, such as official websites, social media platforms, and media broadcasts. Risk assessments, infrastructure resilience plans, public education campaigns, and the implementation of early warning systems must all be included in preparedness plans. By putting these tactics into practice, the detrimental effects of EREs can be mitigated.

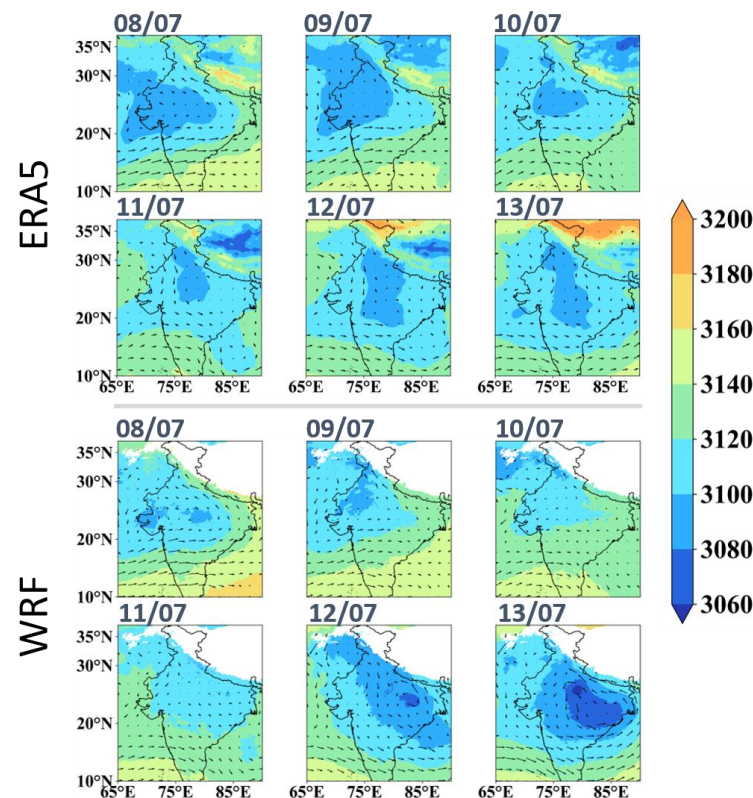


Figure 9. 700 hPa daily geopotential height (m) and wind flow (m/s) from ERA5 (**first** and **second** rows) and WRF model's outer domain simulation (**third** and **fourth** rows) valid for 8–13 July 2023.

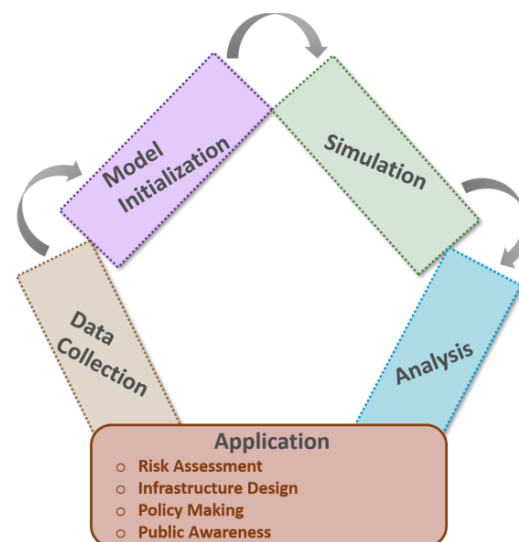


Figure 10. Extreme rainfall events disaster preparedness block diagram.

5. Conclusions

This study comprehensively analyzed an extreme rainfall phenomenon over Himachal Pradesh from 8 to 13 July 2023, using the WRF simulation and reanalysis datasets. The event was characterized by exceptionally high precipitation, with some regions receiving more than 400% of their normal rainfall. A combination of synoptic scale and mesoscale contributed to extreme rainfall, including strong south-westerly moisture transport from the Arabian Sea, interaction with a mid-latitude WD, and intense orographic lifting. The WRF model successfully captured the spatial and temporal evolution of extreme rainfall, with the rainfall peaks aligning well with the observational datasets. However, bias in rainfall magnitude and spatial distribution was evident, necessitating a multi-faceted validation approach incorporating various observational and reanalysis datasets.

The comparison of WRF output with MSWEP and IMD rainfall datasets highlighted the key strengths and limitations of the model. The equitable threat score revealed that WRF exhibited good skill in capturing moderate rainfall thresholds (10–30 mm). However, at the extreme rainfall threshold, the ETS values declined significantly. This indicates that WRF struggled to accurately represent localized heavy rainfall. The MBE analysis suggested that WRF overestimated the rainfall during the peak phase of the event. The correlation analysis confirmed that WRF performed well in capturing broad rainfall patterns during the early phase of the event but struggled in the later stages.

An analysis of VIMT through WRF and ERA5 again exhibited the importance of the large-scale monsoon flow and its interaction with orographic features. Intense south-westerly moisture advection across the Arabian Sea and mid-tropospheric convergence were responsible for continued convection over the region. WRF captured large-scale transport well, but it depicted more substantial localized moisture convergence than ERA5, which was most likely responsible for the higher precipitation estimates. The vertical cross-section of relative humidity (RH) revealed that the event was deep convection driven, as the RH values were above 85%, up to at least 400 hPa, in both the ERA5 and WRF datasets. However, WRF revealed a shallower moist layer, which might indicate that the model slightly underestimated the depth of the convection and possibly the upper-level moisture transport over land.

The results strongly emphasized the importance of a multi-source validation of the simulations for extreme events, especially in complex mountainous terrains like the Western Himalayas, where orographic effects, mesoscale interactions, and synoptic-scale drivers are put together to shape the precipitation pattern. This study underlines the strength of high-resolution WRF simulations in capturing extreme rainfall but also points to areas needing further improvement, especially within convective parameterization, microphysics schemes, and the representation of moisture transport. Future research may explore ensemble-based modeling approaches and advanced data assimilation techniques to enhance model accuracy.

The results have significant implications for sustainable development, particularly in flood forecasting, infrastructure planning, and disaster preparedness. The identification of biases in WRF-simulated extreme rainfall highlights the need for calibrated numerical models in early warning systems to improve lead-time forecasts and enhance disaster response strategies. High-resolution models, when integrated with hydrological risk assessments, can aid in dam management, urban drainage planning, and climate-resilient infrastructure development in landslide-prone regions like Himachal Pradesh. Furthermore, the collaboration between meteorological agencies and disaster management authorities is essential to incorporate real-time numerical simulations into flood risk monitoring systems. Investing in ensemble modeling techniques and machine learning-based bias corrections can enhance model accuracy, leading to better anticipatory planning for extreme weather

events. Additionally, localized adaptation strategies, including watershed management, sustainable land-use planning, and improved flood zoning regulations, can mitigate the socio-economic impacts of extreme rainfall events. Strengthening the integration of high-resolution weather models into disaster risk reduction policies will ultimately support climate resilience and long-term sustainability in vulnerable mountainous regions.

Author Contributions: Conceptualization, N.L. and G.T.; methodology, N.L., G.T., and V.S.; validation, N.L. and S.K.; formal analysis, N.L. and G.T.; data curation, N.L., R.P.G., and S.K.; writing—original draft preparation, N.L., G.T., and R.P.G.; writing—review and editing, G.T.; visualization, N.L., G.T., and V.S. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in the study are included in the article. Further inquiries can be directed to the corresponding author.

Acknowledgments: The authors acknowledge the data-providing agencies, IMD, ECMWF, and MSWEP, for their invaluable contributions to this research. The first author is grateful to the Department of Applied Mathematics, Gautam Buddha University, for the academic and technical support.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

WRF	Weather Research and Forecasting
ECMWF	European Centre for Medium-Range Weather Forecasts
MSWEP	Multi-Source Weighted-Ensemble Precipitation
IMD	India Meteorological Department
ERE	Extreme Rainfall Event
NCEP	National Centers for Environmental Prediction
ISM	Indian Summer Monsoon
ENSO	El Niño–Southern Oscillation
WD	Western Disturbance
FNL	Final Analysis
VIMT	Vertically Integrated Moisture Transport
ETS	Equitable Threat Score
MBE	Mean Bias Error
CC	Correlation Coefficient

References

1. Alifu, H.; Hirabayashi, Y.; Imada, Y.; Shiogama, H. Enhancement of river flooding due to global warming. *Sci. Rep.* **2022**, *12*, 20687. [[CrossRef](#)] [[PubMed](#)]
2. Friess, D.A.; Adame, M.F.; Adams, J.B.; Lovelock, C.E. Mangrove forests under climate change in a 2 °C world. *WIREs Clim. Chang.* **2022**, *13*, 792. [[CrossRef](#)]
3. Goswami, B.N.; Venugopal, V.; Sengupta, D.; Madhusoodanan, M.S.; Xavier, P.K. Increasing Trend of Extreme Rain Events Over India in a Warming Environment. *Science* **2006**, *314*, 1442–1445. [[CrossRef](#)] [[PubMed](#)]
4. Mohapatra, M.; Srivastava, A.K.; Balachandran, S.; Geetha, B. Inter-annual variation and trends in tropical cyclones and monsoon depressions over the north Indian Ocean. *Obs. Clim. Var. Change Over Indian Reg.* **2017**, 89–106. [[CrossRef](#)]
5. Rajeevan, M.; Bhate, J.; Jaswal, A.K. Analysis of variability and trends of extreme rainfall events over India using 104 years of gridded daily rainfall data. *Geophys. Res. Lett.* **2008**, *35*, 035143. [[CrossRef](#)]

6. Guhathakurta, P.; Sreejith, O.P.; Menon, P.A. Impact of climate change on extreme rainfall events and flood risk in India. *J. Earth Syst. Sci.* **2011**, *120*, 359–373. [\[CrossRef\]](#)
7. Gupta, V.; Syed, B.; Pathania, A.; Raaj, S.; Nanda, A.; Awasthi, S.; Shukla, D.P. Hydrometeorological analysis of July-2023 floods in Himachal Pradesh, India. *Nat. Hazards* **2024**, *120*, 7549–7574. [\[CrossRef\]](#)
8. Kumar, P.; Shukla, M.V.; Varma, A.K. Impact of all-sky water vapour channel radiance from INSAT-3D/3DR satellite over South Asia region using WRF model. *Q. J. R. Meteorol. Soc.* **2022**, *148*, 2532–2545. [\[CrossRef\]](#)
9. Pattanaik, D.R.; Rajeevan, M. Variability of extreme rainfall events over India during southwest monsoon season. *Meteorol. Appl.* **2010**, *17*, 88–104. [\[CrossRef\]](#)
10. Roxy, M.K.; Ghosh, S.; Pathak, A.; Athulya, R.; Mujumdar, M.; Murtugudde, R.; Terray, P.; Rajeevan, M. A threefold rise in widespread extreme rain events over central India. *Nat. Commun.* **2017**, *8*, 708. [\[CrossRef\]](#)
11. Fischer, E.M.; Knutti, R. Observed heavy precipitation increase confirms theory and early models. *Nat. Clim. Chang.* **2016**, *6*, 986–991. [\[CrossRef\]](#)
12. Allan, R.P.; Barlow, M.; Byrne, M.P.; Cherchi, A.; Douville, H.; Fowler, H.J.; Gan, T.Y.; Pendergrass, A.G.; Rosenfeld, D.; Swann, A.L.S.; et al. Advances in understanding large-scale responses of the water cycle to climate change. *Ann. N. Y. Acad. Sci.* **2020**, *1472*, 49–75. [\[CrossRef\]](#) [\[PubMed\]](#)
13. Chang, C.; Glover, G.H. Effects of model-based physiological noise correction on default mode network anti-correlations and correlations. *Neuroimage* **2009**, *47*, 1448–1459. [\[CrossRef\]](#)
14. Trenberth, K.E.; Zhang, Y.; Gehne, M. Intermittency in precipitation: Duration, frequency, intensity, and amounts using hourly data. *J. Hydrometeorol.* **2017**, *18*, 1393–1412. [\[CrossRef\]](#)
15. Mishra, V.; Aadhar, S.; Shah, H.; Kumar, R.; Pattanaik, D.R.; Tiwari, A.D. The Kerala flood of 2018: Combined impact of extreme rainfall and reservoir storage. *Hydrol. Earth Syst. Sci. Discuss.* **2018**, *2018*, 1–13.
16. Niu, Z.; Tang, F.; Wang, L. All-sky assimilation of FY-4A AGRI water vapor channels: An observing system experiment study for south Asian monsoon prediction. *Q. J. R. Meteorol. Soc.* **2024**, *150*, 2458–2471. [\[CrossRef\]](#)
17. Schär, C.; Ban, N.; Fischer, E.M.; Rajczak, J.; Schmidli, J.; Frei, C.; Giorgi, F.; Karl, T.R.; Kendon, E.J.; Tank, A.M.G.K.; et al. Percentile indices for assessing changes in heavy precipitation events. *Clim. Change* **2016**, *137*, 201–216. [\[CrossRef\]](#)
18. Yadav, R.K. The recent trends in the Indian summer monsoon rainfall. *Environ. Dev. Sustain.* **2024**, 1–15. [\[CrossRef\]](#)
19. Dash, C.J.; Shrimali, S.S.; Madhu, M.; Kumar, R.; Adhikary, P.P. Unveiling rainfall and erosivity dynamics in Odisha's varied agro-climatic zones for sustainable soil and water conservation planning. *Theor. Appl. Climatol.* **2024**, *155*, 7557–7574. [\[CrossRef\]](#)
20. Singh, R.N.; Sah, S.; Das, B.; Potekar, S.; Chaudhary, A.; Pathak, H. Innovative trend analysis of spatio-temporal variations of rainfall in India during 1901–2019. *Theor. Appl. Clim.* **2021**, *145*, 821–838. [\[CrossRef\]](#)
21. Gadgil, S. The Indian monsoon and its variability. *Annu. Rev. Earth Planet. Sci.* **2003**, *31*, 429–467. [\[CrossRef\]](#)
22. Pattanaik, D.R.; Rathore, L.S.; Kumar, A. Observed and forecasted intraseasonal activity of southwest monsoon rainfall over India during 2010, 2011 and 2012. *Pure Appl. Geophys.* **2013**, *170*, 2305–2328. [\[CrossRef\]](#)
23. Gadgil, S. The monsoon system: Land–sea breeze or the ITCZ? *J. Earth Syst. Sci.* **2018**, *127*, 5. [\[CrossRef\]](#)
24. Wang, Y.; Xu, Y.; Tabari, H.; Wang, J.; Wang, Q.; Song, S.; Hu, Z. Innovative trend analysis of annual and seasonal rainfall in the Yangtze River Delta, eastern China. *Atmos. Res.* **2020**, *231*, 104673. [\[CrossRef\]](#)
25. Dimri, A.; Chevuturi, A.; Niyogi, D.; Thayyen, R.; Ray, K.; Tripathi, S.; Pandey, A.; Mohanty, U. Cloudbursts in Indian Himalayas: A review. *Earth-Sci. Rev.* **2017**, *168*, 1–23. [\[CrossRef\]](#)
26. Hunt, K.M.R.; Fletcher, J.K. The relationship between Indian monsoon rainfall and low-pressure systems. *Clim. Dyn.* **2019**, *53*, 1859–1871. [\[CrossRef\]](#)
27. Dimri, A.; Yasunari, T.; Kotlia, B.; Mohanty, U.; Sikka, D. Indian winter monsoon: Present and past. *Earth-Sci. Rev.* **2016**, *163*, 297–322. [\[CrossRef\]](#)
28. Krishnan, R.; Sundaram, S.; Swapna, P.; Kumar, V.; Ayantika, D.C.; Mujumdar, M. The crucial role of ocean–atmosphere coupling on the Indian monsoon anomalous response during dipole events. *Clim. Dyn.* **2010**, *37*, 1–17. [\[CrossRef\]](#)
29. Hunt, K.M.R.; Turner, A.G.; Stein, T.H.M.; Fletcher, J.K.; Schiemann, R.K.H. Modes of coastal precipitation over southwest India and their relationship with intraseasonal variability. *Q. J. R. Meteorol. Soc.* **2021**, *147*, 181–201. [\[CrossRef\]](#)
30. Martius, O.; Sodemann, H.; Joos, H.; Pfahl, S.; Winschall, A.; Croci-Maspoli, M.; Graf, M.; Madonna, E.; Mueller, B.; Schemm, S.; et al. The role of upper-level dynamics and surface processes for the Pakistan flood of July 2010. *Q. J. R. Meteorol. Soc.* **2012**, *139*, 1780–1797. [\[CrossRef\]](#)
31. Houze, R.A., Jr. Orographic effects on precipitating clouds. *Rev. Geophys.* **2012**, *50*, 1. [\[CrossRef\]](#)
32. Kumar, A.; Asthana, A.; Priyanka, R.S.; Jayangondaperumal, R.; Gupta, A.K.; Bhakuni, S. Assessment of landslide hazards induced by extreme rainfall event in Jammu and Kashmir Himalaya, northwest India. *Geomorphology* **2017**, *284*, 72–87. [\[CrossRef\]](#)
33. Lohan, N.; Routray, A.; Kumar, R.; Mahala, B.K.; Kumar, S. Validation and diagnostic study of cloudburst events over the Himalayan region using IMDAA and ERA5. *Geomat. Nat. Hazards Risk* **2025**, *16*, 2446589. [\[CrossRef\]](#)

34. Allen, K.; Dupuy, J.M.; Gei, M.G.; Hulshof, C.; Medvigy, D.; Pizano, C.; Salgado-Negret, B.; Smith, C.M.; Trierweiler, A.; Van Bloem, S.J.; et al. Will seasonally dry tropical forests be sensitive or resistant to future changes in rainfall regimes? *Environ. Res. Lett.* **2017**, *12*, 023001. [\[CrossRef\]](#)
35. Dobhal, D.P.; Gupta, A.K.; Mehta, M.; Khandelwal, D.D. Kedarnath disaster: Facts and plausible causes. *Curr. Sci.* **2013**, *105*, 00113891.
36. Baidya, S.K.; Shrestha, M.L.; Sheikh, M.M. Trends in daily climatic extremes of temperature and precipitation in Nepal. *J. Hydrol. Meteorol.* **2008**, *5*, 38–51.
37. Vogel, M.M.; Hauser, M.; Seneviratne, S.I. Projected changes in hot, dry and wet extreme events' clusters in CMIP6 multi-model ensemble. *Environ. Res. Lett.* **2020**, *15*, 094021. [\[CrossRef\]](#)
38. Patel, V.; Kuttippurath, J. Significant increase in water vapour over India and Indian Ocean: Implications for tropospheric warming and regional climate forcing. *Sci. Total. Environ.* **2022**, *838*, 155885. [\[CrossRef\]](#)
39. Shukla, B.P.; Kishtawal, C.M.; Pal, P.K. Satellite-based nowcasting of extreme rainfall events over Western Hima-layan region. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2017**, *10*, 1681–1686. [\[CrossRef\]](#)
40. Shah, N.H.; Priamvada, A.; Shukla, B.P. Validation of satellite-based cloudburst alerts: An assessment of location precision over Uttarakhand, India. *J. Earth Syst. Sci.* **2023**, *132*, 161. [\[CrossRef\]](#)
41. Skamarock, W.C.; Klemp, J.B.; Dudhia, J.; Gill, D.O.; Barker, D.M.; Duda, M.G.; Powers, J.G. A description of the advanced research WRF version 3. *NCAR Tech. Note* **2008**, *475*, 10–5065.
42. Chang, H.-I.; Kumar, A.; Niyogi, D.; Mohanty, U.; Chen, F.; Dudhia, J. The role of land surface processes on the mesoscale simulation of the July 26, 2005 heavy rain event over Mumbai, India. *Glob. Planet. Chang.* **2009**, *67*, 87–103. [\[CrossRef\]](#)
43. Chen, C.-S.; Lin, Y.-L.; Peng, W.-C.; Liu, C.-L. Investigation of a heavy rainfall event over southwestern Taiwan associated with a subsynoptic cyclone during the 2003 Mei-Yu season. *Atmos. Res.* **2010**, *95*, 235–254. [\[CrossRef\]](#)
44. Davis, S.; Pentakota, L.; Saptarishy, N.; Mujumdar, P.P. A Flood forecasting framework coupling a high resolution WRF ensemble with an urban hydrologic model. *Front. Earth Sci.* **2022**, *10*, 883842. [\[CrossRef\]](#)
45. Hong, S.-Y.; Lee, J.-W. Assessment of the WRF model in reproducing a flash-flood heavy rainfall event over Korea. *Atmos. Res.* **2009**, *93*, 818–831. [\[CrossRef\]](#)
46. Kadaverugu, R.; Matli, C.; Biniwale, R. Suitability of WRF model for simulating meteorological variables in rural, semi-urban and urban environments of Central India. *Meteorol. Atmos. Phys.* **2021**, *133*, 1379–1393. [\[CrossRef\]](#)
47. Mukhopadhyay, P.; Prasad, V.S.; Krishna RP, M.; Deshpande, M.; Ganai, M.; Tirkey, S.; Sarkar, S.; Goswami, T.; Johnny, C.J.; Roy, K.; et al. Performance of a very high-resolution global forecast system model (GFS T1534) at 12.5 km over the Indian region during the 2016–2017 monsoon seasons. *J. Earth Syst. Sci.* **2019**, *128*, 1–18. [\[CrossRef\]](#)
48. Valappil, V.K.; Temimi, M.; Weston, M.; Fonseca, R.; Nelli, N.R.; Thota, M.; Kumar, K.N. Assessing bias correction methods in support of operational weather forecast in arid environment. *Asia-Pac. J. Atmos. Sci.* **2020**, *56*, 333–347. [\[CrossRef\]](#)
49. Vellore, R.K.; Kaplan, M.L.; Krishnan, R.; Lewis, J.M.; Sabade, S.; Deshpande, N.; Singh, B.B.; Madhura, R.K.; Rama Rao, M.V.S. Mon-soon-extratropical circulation interactions in Himalayan extreme rainfall. *Clim. Dyn.* **2016**, *46*, 3517–3546. [\[CrossRef\]](#)
50. Verma, S.; Panda, J.; Rath, S.S. Role of PBL and microphysical parameterizations during WRF simulated monsoonal heavy rainfall episodes over Mumbai. *Pure Appl. Geophys.* **2021**, *178*, 3673–3702. [\[CrossRef\]](#)
51. Boyaj, A.; Dasari, H.P.; Hoteit, I.; Ashok, K. Increasing heavy rainfall events in south India due to changing land use and land cover. *Q. J. R. Meteorol. Soc.* **2020**, *146*, 3064–3085. [\[CrossRef\]](#)
52. Chawla, I.; Osuri, K.K.; Mujumdar, P.P.; Niyogi, D. Assessment of the Weather Research and Forecasting (WRF) model for simulation of extreme rainfall events in the upper Ganga Basin. *Hydrol. Earth Syst. Sci.* **2018**, *22*, 1095–1117. [\[CrossRef\]](#)
53. Deb, S.K.; Kishitawal, C.M.; Pal, P.K.; Joshi, P.C. Impact of TMI SST on the simulation of a heavy rainfall episode over Mumbai on 26 July 2005. *Mon. Weather. Rev.* **2008**, *136*, 3714–3741. [\[CrossRef\]](#)
54. Kalra, S.; Kumar, S.; Routray, A. Simulation of heavy rainfall event along east coast of India using WRF modeling system: Impact of 3DVAR data assimilation. *Model. Earth Syst. Environ.* **2019**, *5*, 245–256. [\[CrossRef\]](#)
55. Kedia, S.; Vellore, R.K.; Islam, S.; Kaginalkar, A. A study of Himalayan extreme rainfall events using WRF-Chem. *Meteorol. Atmos. Phys.* **2019**, *131*, 1133–1143. [\[CrossRef\]](#)
56. Kumar, A.; Dudhia, J.; Rotunno, R.; Niyogi, D.; Mohanty, U.C. Analysis of the 26 July 2005 heavy rain event over Mumbai, India using the Weather Research and Forecasting (WRF) model. *Q. J. R. Meteorol. Soc.* **2008**, *134*, 1897–1910. [\[CrossRef\]](#)
57. Mohanty, U.C.; Routray, A.; Osuri, K.K.; Prasad, S.K. A study on simulation of heavy rainfall events over indian region with ARW-3DVAR modeling system. *Pure Appl. Geophys.* **2012**, *169*, 381–399. [\[CrossRef\]](#)
58. Mohapatra, G.N.; Rakesh, V.; Ramesh, K.V. Urban extreme rainfall events: Categorical skill of WRF model simulations for localized and non-localized events. *Q. J. R. Meteorol. Soc.* **2017**, *143*, 2340–2351. [\[CrossRef\]](#)
59. Osuri, K.K.; Nadimpalli, R.; Mohanty, U.C.; Niyogi, D. Prediction of rapid intensification of tropical cyclone *Phailin* over the Bay of Bengal using the HWRF modelling system. *Q. J. R. Meteorol. Soc.* **2017**, *143*, 678–690. [\[CrossRef\]](#)

60. Patel, P.; Ghosh, S.; Kagainalkar, A.; Islam, S.; Karmakar, S. Performance evaluation of WRF for extreme flood forecasts in a coastal urban environment. *Atmos. Res.* **2019**, *223*, 39–48. [\[CrossRef\]](#)
61. Routray, A.; Mohanty, U.C.; Niyogi, D.; Rizvi, S.R.H.; Osuri, K.K. Simulation of heavy rainfall events over Indian monsoon region using WRF-3DVAR data assimilation system. *Meteorol. Atmos. Phys.* **2010**, *106*, 107–125. [\[CrossRef\]](#)
62. Vaidya, S.S.; Kulkarni, J.R. Simulation of heavy precipitation over Santacruz, Mumbai on 26 July 2005, using Mesoscale model. *Meteorol. Atmos. Phys.* **2006**, *98*, 55–66. [\[CrossRef\]](#)
63. Chevuturi, A.; Dimri, A.P.; Das, S.; Kumar, A.; Niyogi, D. Numerical simulation of an intense precipitation event over Rudraprayag in the central Himalayas during 13–14 September 2012. *J. Earth Syst. Sci.* **2015**, *124*, 1545–1561. [\[CrossRef\]](#)
64. Kumar, M.S.; Shekhar, M.S.; Krishna, S.S.V.S.R.; Bhutiyani, M.R.; Ganju, A. Numerical simulation of cloud burst event on August 05, 2010, over Leh using WRF mesoscale model. *Nat. Hazards* **2012**, *62*, 1261–1271. [\[CrossRef\]](#)
65. Garg, S.; Tiwari, G.; Azad, S. Evaluation of the WRF model for a heavy rainfall event over the complex mountainous topography of Mandi, India. *Nat. Hazards* **2024**, *120*, 2661–2681. [\[CrossRef\]](#)
66. Karki, R.; Hasson, S.U.; Gerlitz, L.; Schickhoff, U.; Scholten, T.; Böhner, J. Quantifying the added value of convection-permitting climate simulations in complex terrain: A systematic evaluation of WRF over the Himalayas. *Earth Syst. Dyn.* **2017**, *8*, 507–528. [\[CrossRef\]](#)
67. Navale, A.; Singh, C. Topographic sensitivity of WRF-simulated rainfall patterns over the North West Himalayan region. *Atmos. Res.* **2020**, *242*, 105003. [\[CrossRef\]](#)
68. Norris, J.; Carvalho, L.M.; Jones, C.; Cannon, F.; Bookhagen, B.; Palazzi, E.; Tahir, A.A. The spatiotemporal variability of precipitation over the Himalaya: Evaluation of one-year WRF model simulation. *Clim. Dyn.* **2017**, *49*, 2179–2204. [\[CrossRef\]](#)
69. Pai, D.S.; Rajeevan, M.; Sreejith, O.P.; Mukhopadhyay, B.; Satbha, N.S. Development of a new high spatial resolution (0.25×0.25) long period (1901–2010) daily gridded rainfall data set over India and its comparison with existing data sets over the region. *Mausam* **2014**, *65*, 1–18. [\[CrossRef\]](#)
70. Hersbach, H.; Bell, B.; Berrisford, P.; Hirahara, S.; Horányi, A.; Muñoz-Sabater, J.; Nicolas, J.; Peubey, C.; Radu, R.; Schepers, D.; et al. The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* **2020**, *146*, 1999–2049. [\[CrossRef\]](#)
71. Beck, H.E.; Wood, E.F.; Pan, M.; Fisher, C.K.; Miralles, D.G.; Van Dijk, A.I.; Adler, R.F. MSWEP V2 global 3-hourly 0.1 precipitation: Methodology and quantitative assessment. *Bull. Am. Meteorol. Soc.* **2019**, *100*, 473–500. [\[CrossRef\]](#)
72. Giri, R.; Yadav, R.; Ranalkar, M.; Singh, V. Validation of INSAT-3DR sounder retrieved temperature profile with GPS radiosonde and AIRS observations. *Adv. Space Res.* **2022**, *69*, 1100–1115. [\[CrossRef\]](#)
73. Wang, X.; Bi, L.; Wang, H.; Wang, Y.; Han, W.; Shen, X.; Zhang, X. AI-NAOS: An AI-based nonspherical aerosol optical scheme for the chemical weather model GRAPES_Meso5.1/CUACE. *Geosci. Model Dev.* **2025**, *18*, 117–139. [\[CrossRef\]](#)
74. Ullah, K.; Gao, S. Moisture transport over the Arabian Sea associated with summer rainfall over Pakistan in 1994 and 2002. *Adv. Atmos. Sci.* **2012**, *29*, 501–508. [\[CrossRef\]](#)
75. Kalnay, E.C.; Kanamitsu, M.; Kistler, R.; Collins, W.; Deaven, D.; Gandin, L.; Joseph, D. The NMC/NCAR 40-year reanalysis project. *Bull. Am. Meteorol. Soc.* **1996**, *77*, 437–472. [\[CrossRef\]](#)
76. Ganjir, G.; Pattnaik, S.; Trivedi, D. Characteristics of dynamical and thermo-dynamical variables during heavy rainfall events over the Indian region. *Dyn. Atmos. Ocean.* **2022**, *99*, 101315. [\[CrossRef\]](#)

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.